



Access S3 from a VPC

CH

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```
ec2-user@ip-10-0-1-75 ~$ sudo touch /tmp/test.txt
ec2-user@ip-10-0-1-75 ~$ aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-arpita
upload failed: ../../tmp/test.txt to s3://nextwork-vpc-project-arpita/test.txt An error occurred (AccessDenied) when calling the PutObject operation: User:
arn:aws:iam::006834263464:user/NextWork is not authorized to perform: s3:PutObject on resource: "arn:aws:s3:::nextwork-vpc-project-arpita/test.txt" because
no identity-based policy allows the s3:PutObject action
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upload: ../../tmp/test.txt to s3://nextwork-vpc-project-arpita/test.txt
ec2-user@ip-10-0-1-75 ~$ aws s3 ls s3://nextwork-vpc-project-arpita
2025-11-26 19:53:30    2431554 NextWork - Densel is awesome.png
2025-11-26 19:53:31    2399812 NextWork - Lelo is awesome.png
2025-11-26 20:30:10         0 test.txt
ec2-user@ip-10-0-1-75 ~$
```



Introducing Today's Project!

What is Amazon VPC?

Amazon VPC is a Virtual Private Cloud—a logically isolated section of the AWS cloud where you can launch and operate your resources in a secure, customizable network environment. It is useful because it lets you control your networking setup, including IP address ranges, subnets, route tables, security groups, and network gateways, allowing you to design secure and flexible architectures that meet your application's requirements.

How I used Amazon VPC in this project

In today's project, I used Amazon VPC to create a secure and isolated network environment. EC2 instance operates within this private network. The main goal is to establish a direct, private connection between VPC and Amazon S3 using a VPC endpoint. This ensures that traffic to S3 bucket stays within the AWS network, enhancing security and performance. Additionally, I've configured a bucket policy to restrict access to S3 bucket so that only traffic coming through VPC endpoint is allowed.

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One thing I didn't expect in this project was encountering several errors first was finding access key where I had to create a user to find access key. Add permissions AmazonS3ReadOnlyAccess and AmazonS3FullAccess so I can complete remaining project task in the EC2 terminal.

This project took me...

This project took me 2 hours.



In the first part of my project...

Step 1 - Architecture set up

In this step, I will set up a VPC and launch EC2 instance in it because these are the project foundation I need to set up to complete the project.

Step 2 - Connect to my EC2 instance

In this step, I will create an EC2 instance because I want to connect to EC2 instance and try to access a AWS service.

Step 3 - Set up access keys

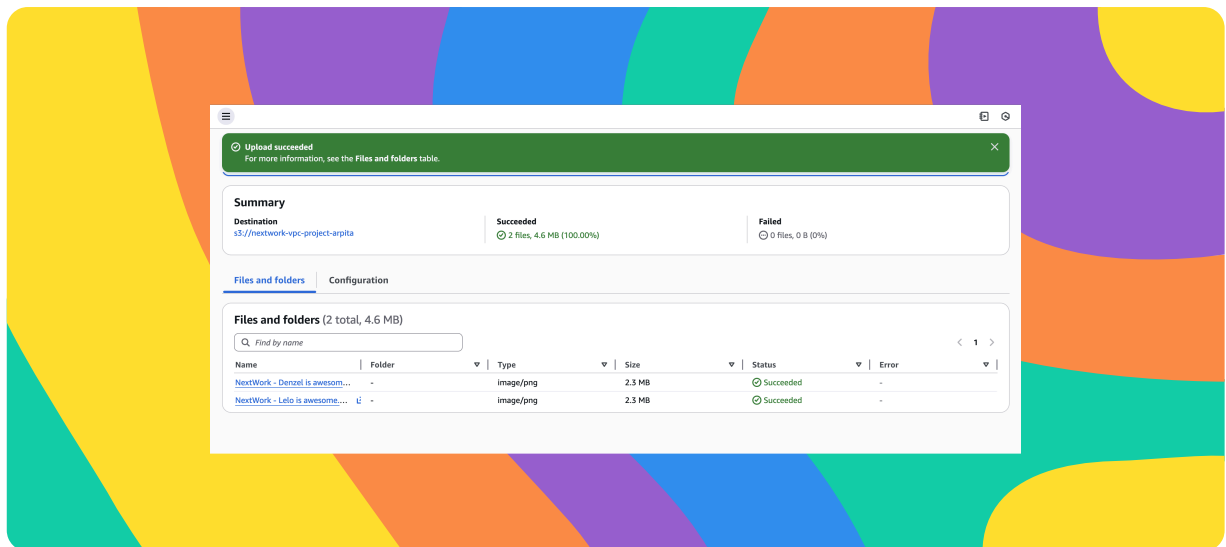
In this step, I will create access keys because give EC2 instance access to AWS environment.



Architecture set up

I started my project by launching EC2 instance.

I also set up the two files from my local computer to S3 bucket.





Access keys

Credentials

To set up my EC2 instance to interact with my AWS environment, I configured the ASW access key, secret access key, the default region name and empty default output format.

Access keys are a set of security credentials consisting of an Access Key ID and a Secret Access Key that allow you to authenticate and make programmatic requests to AWS services using tools like the AWS CLI or AWS SDKs.

Secret access keys are the confidential part of an AWS access key pair that act like a password and are used to securely authenticate programmatic requests to AWS services; they must always be kept private and never shared or exposed.

Best practice

Although I'm using access keys in this project, a best practice alternative is to use an IAM role, which provides temporary, automatically rotated credentials without needing to store or manage access keys manually.



In the second part of my project...

Step 4 - Set up an S3 bucket

In this step, I will create an S3 bucket because I want access it from my EC2 instance and do things like checking what objects are in the bucket.

Step 5 - Connecting to my S3 bucket

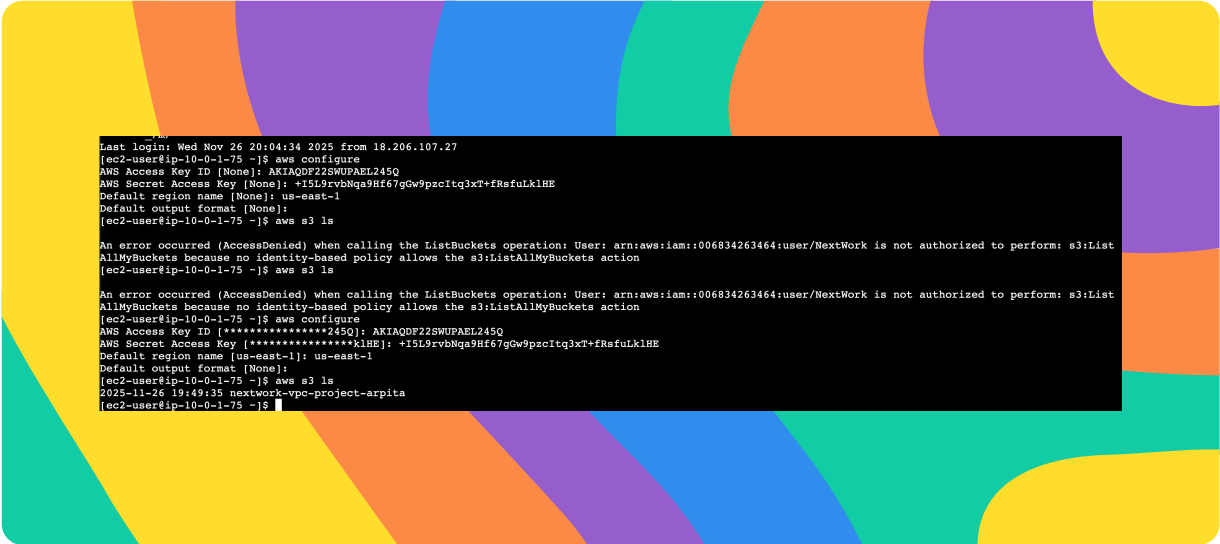
In this step, I will head back to my EC2 instance because I want my EC2 to interact with my S3 bucket.



Connecting to my S3 bucket

The first command I ran was `aws s3 ls`. This command is used to list the S3 buckets in account.

When I ran the command `aws s3 ls` again, the terminal responded with the bucket names. This indicated that my EC2 is interacting with S3 bucket.



```
Last login: Wed Nov 26 20:04:34 2025 from 18.206.107.27
[ec2-user@ip-10-0-1-75 ~]$ aws configure
AWS Access Key ID [None]: AKIAQDF22SWUPAEL245Q
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]:
[ec2-user@ip-10-0-1-75 ~]$ aws s3 ls

An error occurred (AccessDenied) when calling the ListBuckets operation: User: arn:aws:iam::006834263464:user/NextWork is not authorized to perform: s3:ListAllMyBuckets because no identity-based policy allows the s3:ListAllMyBuckets action
[ec2-user@ip-10-0-1-75 ~]$ aws s3 ls

An error occurred (AccessDenied) when calling the ListBuckets operation: User: arn:aws:iam::006834263464:user/NextWork is not authorized to perform: s3:ListAllMyBuckets because no identity-based policy allows the s3:ListAllMyBuckets action
[ec2-user@ip-10-0-1-75 ~]$ aws configure
AWS Access Key ID [*****245Q]: AKIAQDF22SWUPAEL245Q
AWS Secret Access Key [*****klHE]: 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
Default region name [us-east-1]: us-east-1
Default output format [None]:
[ec2-user@ip-10-0-1-75 ~]$ aws s3 ls
2025-11-26 19:49:35 nextwork-vpc-project-arpita
[ec2-user@ip-10-0-1-75 ~]$
```



Connecting to my S3 bucket

Another CLI command I ran was `aws s3 ls s3://nextwork-vpc-project-arpita` which returned the files I uploaded from my local computer to S3 bucket.

```
Last login: Wed Nov 26 20:04:34 2025 from 18.206.107.27
[ec2-user@ip-10-0-1-75 ~]$ aws configure
AWS Access Key ID [None]: AKIAQDF22SMUPAEL245Q
AWS Secret Access Key [None]: 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
Default region name [None]: us-east-1
Default output format [None]:
[ec2-user@ip-10-0-1-75 ~]$ aws s3 ls

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Default region name [us-east-1]: us-east-1
Default output format [None]:
[ec2-user@ip-10-0-1-75 ~]$ aws s3 ls
2025-11-26 19:49:35 nextwork-vpc-project-arpita
[ec2-user@ip-10-0-1-75 ~]$ aws s3 ls s3://nextwork-vpc-project-arpita
2025-11-26 19:53:30      2431554 NextWork - Denzel is awesome.png
2025-11-26 19:53:31      2399812 NextWork - Lolo is awesome.png
[ec2-user@ip-10-0-1-75 ~]$
```



Uploading objects to S3

To upload a new file to my bucket, I first ran the command `sudo touch /tmp/test.txt`. This command creates a blank .txt file in EC2 instance.

The second command I ran was `aws s3 cp /tmp/test.txt s3://nextwork-vpc-project-arpita`. This command is broken down here: `aws s3 cp`: is the command to copy files. `/tmp/test.txt` is the source file path. It's saying that the file to copy is `test.txt`, which exists inside a folder called `tmp`. `s3://nextwork-vpc-project-yourname` is the destination path. It's saying that the file should be copied to the S3 bucket `vpc-project-yourname`!

The third command I ran was `aws s3 ls s3://nextwork-vpc-project-arpita` which validated that object is created and shows what files are present in the new S3 bucket.

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